PID-Guard: Detecting Prompt Injection Attacks in Large Language Models Using Semantic Intent Drift and Behavioral Analysis

1. Introduction

PID-Guard is an AI-powered platform designed to detect and verify whether a given piece of text is human-written or generated by artificial intelligence. With the rapid rise of large language models, there is an increasing risk of misuse in areas such as academic dishonesty, misinformation, phishing attacks, and synthetic content generation.

Existing detection tools are either inaccessible to the public, lack transparency, or fail against advanced AI models. PID-Guard addresses these limitations by integrating statistical analysis, machine learning techniques, and explainable AI to provide accurate, real-time detection results.

The system combines token probability analysis, model-based classification, and explainability layers to deliver a user-friendly and trustworthy detection framework. This sprint focuses on building the core detection pipeline and scoring mechanism.

2. Product Goal

The primary goal of this sprint is to develop a reliable and accessible AI-generated text detection system that enables users to verify content authenticity.

The system must ensure:

- Input text is analyzed using statistical and machine learning models
- AI-generated patterns are detected with high accuracy
- A confidence score is generated indicating likelihood of AI generation
- Results are explained in simple, human-understandable language
- The system is fast, scalable, and user-friendly

This aligns with the broader objective of promoting **trust, transparency, and responsible AI usage**.

3. Demography (Users, Location)

Users

Target Users:

- Students and educators verifying academic content
- Journalists and fact-checkers validating articles
- Organizations detecting AI-generated phishing or fraud content
- General users verifying authenticity of online text

User Characteristics:

- Mixed technical background — system must be simple and intuitive
- Need quick and reliable results
- Require transparent explanations for decisions
- Prefer accessible, web-based tools

Location

Target Location: Global deployment. PID-Guard is designed as a web-based platform accessible from anywhere, supporting users across different regions without dependency on local infrastructure.

4. Business Processes

The key business processes supported by PID-Guard are:

Text Submission and Processing

Users submit text through the interface. The system preprocesses the text by cleaning, tokenizing, and preparing it for analysis.

AI Detection and Scoring

The processed text is analyzed using detection models (e.g., statistical token analysis, ML classifiers). A probability score (0–100) is generated indicating AI likelihood.

Explainability Generation

An explanation module generates a human-readable summary explaining why the content is classified as AI-generated or human-written.

Result Delivery and Visualization

The system highlights suspicious sections and displays results with confidence scores and explanations.

Logging and Monitoring

All interactions are logged for auditing, performance monitoring, and system improvement.

5. Features

AI Text Detection Engine

Description:

The system analyzes input text using statistical and machine learning models to determine whether it is AI-generated. It identifies patterns such as repetition, predictability, and token probability distribution.

User Story:

As a user, I want to know whether a piece of text is AI-generated so that I can verify its authenticity.

Confidence Scoring System

Description:

The platform assigns a score (0–100) representing the likelihood that the text is AI-generated. Higher scores indicate stronger AI probability.

User Story:

As a user, I want a clear score so that I can quickly understand how likely the text is AI-generated.

Explainable AI Module

Description:

The system provides a natural language explanation of the detection result, highlighting suspicious patterns and reasoning behind the classification.

User Story:

As a user, I want to understand why the system flagged the text so that I can trust the result.

Highlighted Text Analysis

Description:

The system highlights specific parts of the text that appear AI-generated or suspicious, providing deeper insights.

User Story:

As a user, I want to see which parts of the text are flagged so that I can analyze them more closely.

Logging and Monitoring System

Description:

All inputs, outputs, and system decisions are recorded for auditing and system improvement purposes.

User Story:

As an administrator, I want to track system activity so that I can monitor performance and detect misuse.

6. Authorization Matrix

The following table defines the roles within PID-Guard and their corresponding access levels:

Role	Access Level	Permissions
User	Restricted	Submit text for analysis; view detection results

Role	Access Level	Permissions
Researcher	Moderate	Analyze results; access detailed insights and logs
Admin	Elevated	Monitor system activity; manage users; view all logs
System Administrator	Full	Full access to system configuration, models, and infrastructure

7. Assumptions

- The detection models (statistical and ML-based) will be available and functional
- The system will be deployed as a web application accessible via browsers
- Users will input text in supported formats (plain text)
- The explanation module will use a lightweight LLM (e.g., Mistral or similar)
- Logging and monitoring infrastructure will be available
- The development team has access to Python, NLP libraries, and ML frameworks
- Stakeholders will provide feedback during testing phases
- Advanced multimodal detection (image/audio) is out of scope for this phase